

AXIS 0

Zero-boundary combinatorial memory palace

Greatly extended formalization of Tom Klootwijk’s corpus

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The enhanced definition

A spatial, field-defined knowledge system whose rooms identify combinations of **microstructures**, **chemical geometries** and **bacterial compositions**. Operators, expressions, observations and their provenance share explicit field encodings. Zero-boundary queries identify interfaces; a separate witness calculus identifies failed model conditions without discarding the underlying proposal.

Central extension. The source’s “0 day” is treated as its biomimicry boundary-state concept, not the chemical or software-exploit interpretation that the author explicitly corrected.

Verification. Exact definitions, proofs, counterexamples and an executed integer reference accompany a 45-room synthetic memory palace. The verification record separates a proved representation property, a tested computation and an empirically supported biological claim.

1 Controlling meaning and the enhanced system

SOURCE INSTRUCTIONS / RETAINED INTENT

Your clarification in the new document asks for “0 day nature biomimicry mechanisms defined as 0 of the signed distance field” and a joint effort to map their combinatorics in a spatial memory palace. You then specify “microstructures chemical geometries and bacterial compositions” and bring the discussion back to your Axis 0 corpus. [S4, pp. 3, 6, 8]

The opening chemical/software interpretation in the export is therefore not the controlling definition. This revision follows the author’s correction and retains the three-layer structure developed afterward.

One domain; several explicit roles

Let

$$\mathcal{A}_0 = \text{Closure}_{\Gamma}(\text{FieldEnc}(\Sigma)) \quad (1.1)$$

be the domain of all operator-fields admitted by a declared vocabulary Σ and constructors Γ . The 0 in *Axis 0* names ALL admitted operators; it is not a physical coordinate, a zero-size memory budget or a claim that all distance values vanish. [S1, p. 5]

For an address domain X , retain Boolean double $\mathbb{U}\mathbb{U}$:

$$\mathbb{U}_X = \mathbb{B}^X, \quad \mathbb{B} = \{0, 1\}, \quad \mathbb{U}\mathbb{U}_X = \mathbb{U}_X \times \mathbb{U}_X = (U_F, U_O). \quad (1.2)$$

The roles U_F and U_O are specified by the chosen transition profile. The three biological-description layers are additional *typed contents*, not three new meanings silently substituted for those two roles.

The complete palace record

A realization is

$$\mathcal{P}_{\Theta} = (\mathcal{A}_0, \mathcal{C}, \mathcal{H}, \mathcal{L}, \Psi, \mathcal{W}, \mathcal{D}), \quad (1.3)$$

where $\mathcal{C}$ is the versioned catalogue product, $\mathcal{H}$ the spatial/hierarchical address structure, $\mathcal{L}$ the provenance ledger, Ψ the field transition, $\mathcal{W}$ the witness observer, and $\mathcal{D}$ the readout. The profile Θ assigns metrics, units, data layouts, predicates and transition rules.

What this adds

A typed microstructure--chemical--composition record; exact spatial and canonical SDF models; injective room addressing; an integer composition algebra; a proposal-aware bug finder; multi-label conflict witnesses; temporal snapshots; and a reproducible verification package.

2 Source ledger: what is preserved and what is verified

NO SILENT SUBSTITUTIONS BETWEEN AUTHOR INSTRUCTIONS AND EXPORTED EXPLANATIONS

The uploaded file contains nine content-bearing pages followed by a blank tenth page. References here use its original content-page numbers. Pages 4--5 contain the room/corridor/outer-palace table; page 6 contains the three-layer pipeline; pages 7--8 give proposed coupling formulas; page 9 attempts the Axis 0 reinterpretation.

Source content	Version 6 treatment
Author: zero-day as SDF zero-boundary biomimicry; spatial memory palace.	Controlling task definition; formalized as an interface-event query. [S4, p. 3]
Author: microstructures, chemical geometries, bacterial compositions.	Retained as μ, χ, β layers with distinct types and units. [S4, p. 6]
Rooms, corridors, unions and padding.	Given separate address, relation, set-operation and offset semantics. [S4, pp. 4--5]
Eikonal constraint prevents memory corruption.	Verified as a geometric condition only; address injectivity is proved independently. [S4, p. 5]
Directional gradient zero locates molecular equilibrium.	A directional derivative alone is insufficient; the missing energy/constraint law is an explicit parameter. [S4, p. 7]
Monotone frontier evolution prevents all spatial overlap.	Single-field monotonicity is proved; label collisions are detected by a separate observer. [S4, p. 9]
Infinite resolution without extra memory.	Distinguished from storing arbitrary independent detail or enumerating every combination. [S4, p. 8]

Statement categories

S / Source: an author's instruction or an attributed statement in an export. **D / Definition:** an explicit construction introduced in this revision. **P / Proof:** a conclusion derived from stated definitions. **V / Verified execution:** a result of the included finite tests. **E / Empirical:** a measured biological or chemical claim with its own evidence record.

No measured biological data accompany the new source. Its named natural mechanisms can therefore enter as source descriptions; a calculated geometric contact is not promoted to an experimentally established material property.

The core results below are constructive mathematics and executed reference semantics. Their conditions are part of the formal definition, not hidden changes to the requested behavior.

3 Typed fields across the three layers

DEFINITION / ONE ENCODING DOES NOT ERASE SCIENTIFIC TYPE

For a room r , define

$$\mathcal{R}_r = (r, \mu_r, \chi_r, \beta_r, \Theta_r, \mathcal{I}_r, \mathcal{L}_r). \quad (3.1)$$

The address r identifies the combination, Θ_r describes its metric and interpretation, $\mathcal{I}_r$ contains its interaction predicates, and $\mathcal{L}_r$ links its evidence and revisions.

Layer	Data type	Required interpretation
μ	Geometry, phase, boundary and optional distance.	Is the occupied region material, a cavity, or an envelope? What are the metric and units?
χ	Labeled molecular graph, geometry record, annotation fields or vector-valued descriptors.	Which values are distances, labels, charges, energies or concentrations? Which source fixes them?
β	Component identities, integer counts or rational fractions, occupancy/support and time labels.	Which entities are counted, what is the denominator, and which region/time does the observation describe?

A common coordinate query is a typed tuple

At an address x and snapshot n , a room query can return

$$Q_r(x, n) = (d_{\mu,r}(x), \chi_r(x), n_{\beta,r}(x, n), \mathcal{I}_r(x, n)). \quad (3.2)$$

A channel is called an SDF only when its magnitude is distance to its declared zero boundary under a specified metric. A label or density channel can *have an SDF encoding* without becoming a physical distance quantity itself.

The same distinction applies to operators. Their encoded field is read as a rule through an assigned decoder; geometric shape alone is not an unspecified instruction set.

Units and missing values are first-class data

A dimensional field records its unit and scale. Different units are not added merely because their channels share a coordinate. A missing composition, unknown interaction coefficient or unmeasured material property is represented as **unknown**, not as numerical zero.

Three zeros, three roles

Axis 0: ALL admitted operators. $d = 0$: a boundary. $n_k = 0$: zero recorded count of a component. Their notation can coexist without identifying their meanings.

[S4, three-layer pipeline p. 6 and channels p. 7; typing introduced here]

4 All definitions as exact canonical SDFs

DEFINITION AND PROOF / THE INFORMATION CARRIER

A finite description---a trait record, operator table, expression, ledger assertion or time rule---is serialized into a declared bitword $b = (b_i)$. Fix an address graph with metric d_G and a normal coordinate $z \in \mathbb{R}$. Define

$$s_i = 2b_i - 1, \quad D((i, z), (j, w)) = 4d_G(i, j) + |z - w|. \quad (4.1)$$

For connected distinct addresses, $d_G(i, j) \geq 1$. The associated occupied set and field are

$$\Omega_b = \{(i, z) : z \leq s_i\}, \quad \boxed{d_b(i, z) = z - s_i = z + 1 - 2b_i}. \quad (4.2)$$

P1 / Exactness of the canonical distance

The boundary on fiber i is (i, s_i) . A boundary on another fiber j cannot be closer, because $|s_i - s_j| \leq 2$ and

$$4d_G(i, j) + |z - s_j| \geq |z - s_i| + 2 \quad (j \neq i). \quad (4.3)$$

Thus $|d_b(i, z)|$ is exactly the nearest-boundary distance. Its sign is negative inside Ω_b , positive outside, and zero at the boundary. In particular,

$$\boxed{b_i = \mathbf{1}\{d_b(i, 0) < 0\}}. \quad (4.4)$$

This gives a lossless common field encoding of every finite description, with the bitword length and decoder declared. It is a construction of this formalization, not a claim that one sign bit stores every possible original Euclidean distance.

Operator closure

If $\text{Enc}(p)$ is a field encoding of a rule description and $\text{Enc}(x)$ encodes its operands, define

$$\text{App}(\text{Enc}(p), \text{Enc}(x)) = \text{Enc}(\text{Eval}_\Theta(p, x)) \quad (4.5)$$

whenever the evaluation returns. The result is again an encoded field in $\mathcal{A}_0$. The bootstrap reader and admitted constructors have their own descriptions in this language; their interpretations remain explicit.

The *canonical SDF* encodes information. The *spatial SDF* describes a modeled room or interface. The *empirical mechanism* is the biological/chemical object a record refers to. Encoding the first does not automatically validate the other two.

5 The zero-boundary and exact spatial graph distance

SOURCE DEFINITION MADE OPERATIONAL / INTEGER HALF-EDGE METRIC

Let $G = (V, E)$ be a finite undirected graph with unit-length edges and phase $F : V \rightarrow \mathbb{B}$. Give each edge a geometric line segment. The source's midpoint boundary is

$$Z_F = \{\text{mid}(u, v) : \{u, v\} \in E, F(u) \oplus F(v) = 1\}. \quad (5.1)$$

This retains the exact formula in the last page of the new source. It places the zero on the edge interface, not at an arbitrary active vertex. [S4, p. 9]

Define the interior along each edge by its endpoint phases: a uniform edge stays in that phase; an unlike-phase edge changes at its midpoint. The metric is shortest path length on this geometric graph. At a vertex,

$$d_F(v) = (1 - 2F(v)) \text{dist}_G(v, Z_F). \quad (5.2)$$

For an edge-connected component without an interface, the distance is *unassigned/no boundary*. The reference code returns `None`; it does not manufacture a finite surface distance. No exterior ghost cells are added unless the profile declares them.

P2 / Integer reconstruction

Let B_F be the set of endpoints of unlike-phase edges. Then, where a boundary is reachable,

$$\text{dist}_G(v, Z_F) = \frac{1}{2} + \min_{u \in B_F} \text{dist}_G(v, u). \quad (5.3)$$

Every path to a midpoint enters its edge from an endpoint and then travels half an edge. Minimizing over all such endpoints gives the equation. Therefore

$$d_F^{(2)}(v) := 2d_F(v) \quad (5.4)$$

is an odd signed integer. A multi-source wave initialized at B_F with magnitude 1 increments magnitude by 2 per graph layer and reconstructs it exactly.

Present as a queryable interface

The source's biomimetic “0-day” is formalized as

$$\text{Present}(F, n) = \{(n, e) : e \in Z_{F_n}\}. \quad (5.5)$$

It identifies a boundary at a snapshot. It is not a software-exploitation procedure or a discovery-time claim. There is no numerical tolerance hidden in the midpoint definition.

[General distance/SDF distinction: R1. Graph reconstruction and exact integer convention: new construction.]

6 Layer μ : microstructure, cavity and metric

STRUCTURAL DESCRIPTION / GEOMETRY AND MECHANISM ARE SEPARATE RECORDS

A microstructure entry has the form

$$\mu = (\text{id}, \text{role}, G, F_\mu, Z_\mu, d_\mu, \text{units}, \text{parameters}, \text{provenance}). \quad (6.1)$$

The role is explicitly **solid**, **cavity** or **envelope**. This resolves the source's use of $f_\mu < 0$ both for physical material and for an interior micro-cavity. The chosen role fixes which interpretation applies in a room. [S4, pp. 6--7]

In the worked profile, $F_\mu = 1$ means **inside a modeled chamber**. It does not mean bacterial biomass or a measured functional material. Chemical annotations and composition support may occupy the same chamber addresses by design.

Exact transformations and units

For an exact Euclidean SDF d and similarity $x' = a + sRx$, with $s > 0$ and R orthogonal, the exact transformed field is

$$d'(x') = s d(R^\top(x' - a)/s). \quad (6.2)$$

The output factor s follows from scaling all distances by s . A spatial log-polar address is not itself this geometric transform; the two roles require separate maps.

For a fixed binary grid, a coordinate shift moves phase samples. Changing the samples changes the boundary and may require recomputing $d_F^{(2)}$. Retaining an old magnitude array after arbitrary bit flips need not give the new signed distance.

The source's Eikonal statement

For a differentiable exact Euclidean SDF, the local condition $\|\nabla d\| = 1$ describes its distance behavior. It says nothing about whether two room records were assigned the same address. For example, $d(x) = x$ satisfies the condition, while an independently chosen address function $h(r) = 0$ collides for every pair of distinct room ids.

The discrete profile uses a different, testable metric condition: adjacent vertex readings satisfy $|d_F^{(2)}(u) - d_F^{(2)}(v)| \leq 2$. A classical Euclidean gradient is not assigned to an un-interpolated bitplane.

Geometric regularity, correct room addressing, and biological function are distinct properties. Each has its own field data and verification condition. The source's Eikonal equation is retained in its geometric role rather than used as a memory-integrity theorem.

[S4, pp. 2, 5; transformation and address counterexample derived here]

7 Layer chi: chemical graphs and geometric annotations

TYPED CHEMICAL DESCRIPTION / NO SUBSTITUTION OF SHAPE FOR EQUILIBRIUM

Use the record

$$\chi = (\text{id}, G_\chi, \ell_V, \ell_E, \mathbf{q}, \mathbf{a}, \text{units}, \text{provenance}), \quad (7.1)$$

where G_χ stores connectivity, ℓ_V and ℓ_E label vertices and edges, $\mathbf{q}$ optionally stores positions, and $\mathbf{a}$ supplies additional scalar/vector annotations. Labeled atom vertices and bond edges agree with IUPAC's molecular-graph definition. [R2]

An annotation vector such as the source's

$$\mathbf{f}_\chi = (f_{\text{alkane}}, f_{\text{benzene}}, f_{\text{halogen}})$$

is preserved as a possible channel bundle. Each component still needs a meaning and unit. The presence of a named channel does not specify its energy, concentration, occupancy or distance formula. [S4, p. 7]

When chemical geometry has a zero boundary

A declared geometric support Ω_χ can have its own phase and SDF. A graph can also receive a metric or embedding. Connectivity alone does not select unique coordinates, bond lengths or stereochemistry. The source does not provide those measurements; the model stores them when supplied rather than deducing them from the label.

The directional-gradient condition is not an equilibrium law

For scalar f_χ , the condition $\nabla f_\chi \cdot v = 0$ says that the directional derivative along v vanishes. Take

$$f_\chi(x, y) = x, \quad v = (0, 1). \quad (7.2)$$

Then $\nabla f_\chi = (1, 0) \neq 0$ but $\nabla f_\chi \cdot v = 0$ everywhere. Thus the source expression cannot alone identify a chemical equilibrium.

A chosen energy model $E(q)$ would instead require its own equilibrium conditions, such as $\nabla E = 0$ for an unconstrained stationary point, with further conditions to classify stability. For vector-valued $\mathbf{f}_\chi$, the directional derivative is $D\mathbf{f}_\chi v$. Neither an energy model nor its parameters are supplied by the source equation.

Executable interpretation used in the example

The example's χ_0, χ_1, χ_2 are labeled annotation footprints. Their geometric containment can be checked exactly. Their labels do not assert a measured affinity, a molecular synthesis pathway or a specific biological effect.

8 Layer beta: composition is not a one-bit concentration

INTEGER AND RATIONAL FIELDS / EXACT THRESHOLDING

A composition entry specifies component identities and nonnegative integer counts

$$\mathbf{n}(x, n) = (n_1, \dots, n_K) \in \mathbb{N}^K. \quad (8.1)$$

For a declared total $Q > 0$, normalized fractions are $p_k = n_k/Q$ when $\sum_k n_k = Q$. Then $\mathbf{p}$ lies in the probability simplex. At zero total, the record denotes absence; it is not normalized by 0/0.

The bacterial layer may contain biological species names when source records supply them, but the numerical operations depend on explicit identities, counts, domains and observation times. The reference examples use anonymous component labels and synthetic counts.

Bitplane representation of finite precision

For $0 \leq n_k(x) < 2^b$, represent

$$n_k(x) = \sum_{j=0}^{b-1} 2^j B_{k,j}(x), \quad B_{k,j} \in \mathbb{B}^X. \quad (8.2)$$

There is one bit per cell *per precision plane*. This is exact integer storage; it does not fit every density into a single bit. The occupancy and threshold fields are derived observations:

$$F_{\beta,\tau}(x) = \mathbf{1}\{n_1(x) + \dots + n_K(x) \geq \tau\}. \quad (8.3)$$

Their zero boundaries are reconstructed from those masks. The underlying density/count field remains a different object.

The source's depth-derived concentration formula

For a cavity-role SDF, the proposed

$$\rho(x) = \min\{1, \max\{0, -c d_\mu(x)\}\}, \quad c \geq 0, \quad (8.4)$$

is a bounded *model assignment*. For $c > 0$ and $0 < \tau \leq 1$, its threshold region is exactly $d_\mu(x) \leq -\tau/c$. This follows by monotonicity of clipping. It defines a depth profile, not a growth equation, transport law or measured bacterial concentration. [S4, p. 7]

A uniform composition can have no internal concentration boundary. A valid three-layer room therefore need not contain a point where three channels all vanish. The worked room passes its declared predicates while its triple-zero intersection is empty.

9 A temporal composition law with a conservation proof

NEW OPTIONAL PROFILE / DIGITAL INVENTORY TRANSPORT

The source describes density and growth gradients but does not supply a rate law. A concrete dynamics can still be defined at the bookkeeping level without assigning unprovided biological parameters.

Let $J_{k,n}(u, v) \in \mathbb{N}$ be an explicitly supplied transfer of component k from cell u to cell v during a step. Use the previous snapshot to define

$$n_{k,n+1}(v) = n_{k,n}(v) + \sum_u J_{k,n}(u, v) - \sum_w J_{k,n}(v, w). \quad (9.1)$$

The transfer profile states its allowed edges and imposes the inventory condition

$$\sum_w J_{k,n}(v, w) \leq n_{k,n}(v). \quad (9.2)$$

This is the domain of a conservative transfer, not an unstated automatic change to a user's proposal. A proposal outside it has a specific inventory witness.

P3 / Nonnegativity and conservation

Subtracting the outgoing amount leaves a nonnegative remainder; adding incoming transfers preserves it. Summing over all cells cancels every internal transfer once positively and once negatively:

$$\sum_v n_{k,n+1}(v) = \sum_v n_{k,n}(v). \quad (9.3)$$

The conclusion holds component by component. A production, loss or exchange with an external reservoir requires an additional declared source term and changes the balance equation accordingly.

Why this is different from the monotone frontier

The visited field in a wave can only grow, but local component counts can both increase and decrease under transport. A decrease caused by a recorded transfer is not automatically “phase corruption.” One represents explored locations; the other represents material inventory in the chosen digital model.

Verified execution

The reference tests enumerate 2,000 nonnegative count/transfer configurations on a three-cell chain. Every result preserves the total and agrees with a direct scalar update. This verifies the supplied arithmetic law; a real biological transfer rate would need a separate observation or model definition.

10 All combinatorics as a versioned room product

DEFINITION AND PROOF / COLLISION-FREE LOGICAL ROOM IDENTITY

Let the catalogues at version v be

$$C_\mu^{(v)}, \quad C_\chi^{(v)}, \quad C_\beta^{(v)}, \quad |C_\mu| = m, \quad |C_\chi| = c, \quad |C_\beta| = b. \quad (10.1)$$

The candidate palace is the Cartesian product

$$C^{(v)} = C_\mu^{(v)} \times C_\chi^{(v)} \times C_\beta^{(v)}. \quad (10.2)$$

Each triple (i, j, k) has the integer room index

$$q = ((i \, c) + j)b + k. \quad (10.3)$$

The full identity includes the catalogue version: (v, q) . Changing catalogue sizes without changing the version would change the interpretation of old indices.

P4 / Exact inverse and uniqueness

Recover (i, j, k) by

$$k = q \bmod b, \quad t = \lfloor q/b \rfloor, \quad j = t \bmod c, \quad i = \lfloor t/c \rfloor. \quad (10.4)$$

Substitution recovers the original bounded tuple. Therefore different triples have different indices, and every integer $0 \leq q < mcb$ has exactly one triple. This is an address-injectivity proof, separate from SDF smoothness or field monotonicity.

The accepted subspace

Let $P_1, \dots, P_s$ be explicitly declared predicates on a room. Then

$$C_{\text{pass}} = \{r \in C : \forall h, P_h(r) = 1\}. \quad (10.5)$$

A failed predicate returns its witness. A predicate whose inputs are unknown remains unknown. The product contains candidate combinations; passing a synthetic geometric rule is not silently renamed biological compatibility.

Finite enumeration and factorized storage

The product can be stored by catalogue references without materializing all mcb rooms. Enumerating every room or storing one independent result per room still has that cardinality. For three catalogues of 1,024 entries each, even *one* result bit per triple occupies 2^{30} bits, or 128 MiB. Thirty-two such channels occupy 4 GiB, before geometry or provenance.

The mathematical index space can grow without bound through successive finite versions. That is different from holding infinitely many independent room states in finite VRAM.

11 Spatial placement, octrees and log-polar addresses

THREE INDEPENDENT COORDINATE ROLES / NO CONFLATION WITH GEOMETRY

A room index identifies a tuple. A palace location organizes retrieval. A physical coordinate describes a modeled mechanism. Specify each map separately:

$$(v, q) \xrightarrow{l} \text{palace address} \quad \text{and} \quad \xi_r : X_r \rightarrow \text{declared geometric carrier}. \quad (11.1)$$

An octree-compatible integer placement

For bounded integer coordinates $x, y, z < 2^d$, define the Morton index

$$M_d(x, y, z) = \sum_{h=0}^{d-1} (x_h 2^{3h} + y_h 2^{3h+1} + z_h 2^{3h+2}). \quad (11.2)$$

Bits in separate residue classes modulo 3 reconstruct the three coordinates, proving uniqueness at fixed depth d . Depth is part of the address; an optional leading sentinel 2^{3d} can record it. The code tests 37,449 round trips over depths 0 through 5.

For actual disjoint room cells, use an injective placement into half-open cubes $a + [0, L)^3$ on an L -spaced lattice. Their interiors do not overlap by construction. A corridor is an explicit edge or geometric connector between selected ports; it is not automatically the CSG intersection of two disjoint rooms.

Log-polar definition storage

A record index ρ and bit offset j can be given the retained log-polar labels

$$\log_{\varphi}(r/r_0) = \rho/B, \quad \theta/(2\pi) = (j + \tfrac{1}{2})/J. \quad (11.3)$$

The selected scale $\varphi > 1$, bins B, J and origin convention belong to the profile. These addresses require no trigonometry for field lookup. They do not make the canonical information metric Euclidean polar distance.

Retaining tree and parity structure

For a branch bit $c_n = q_n \oplus j_n$, retain $a_{n+1} = (a_n \ll 1) \mid c_n$ with root sentinel $a_0 = 1$. The jitter rule j_n is declared. An ordered search additionally requires ordered keys; a jittered branch walk is not automatically a binary search.

Klein-carrier parity remains an optional chart rule: $d_j = (-1)^h d_i$ and $p_j = p_i \oplus h$. The lifted reading $(-1)^p d$ is consistent. A chemical label or a component count does not acquire a minus sign merely because a spatial chart reverses orientation. [S1–S2; typed integration]

12 Operators as field-defined expressions

EXECUTABLE REFERENCE / RULES ARE ENCODED DATA

Each named rule is a finite record containing its name, truth field and self-reference. The reference library uses three-input truth tables, indexed by $a + 2b + 4c$:

$$\text{Gate}_t(a, b, c) = t[a + 2b + 4c], \quad 0 \leq t < 256. \quad (12.1)$$

A rule record is serialized and encoded by the canonical SDF construction in Section 4, exactly like room data or a witness report. A common reader evaluates its eight rows; it does not infer the rule from an unlabeled geometric outline.

Field name	Table	Meaning in the reference profile
OR	0xEE	$a \vee b$.
AND	0x88	$a \wedge b$.
XOR	0x66	$a \oplus b$.
AND_NOT	0x22	$a \wedge \neg b$.
FRESH	0x08	$a \wedge b \wedge \neg c$.
ALL_THREE	0x80	$a \wedge b \wedge c$.

The palace computation as a graph of fields

A query is a composition of Quote, Decode, Boundary, Threshold, Rank, Gate, Witness and Readout operators. Each has an encoded description and declared input/output types. Integer counts use as many bitplanes as their specified precision requires.

For the worked catalogue, three 45-bit predicate fields are composed by ALL_THREE. The resulting 45-bit room field identifies exactly the passing triples. The scalar witness records retain the reason for every excluded candidate; the bitwise result and record-by-record evaluation are checked to agree.

Closure is not one-operation completeness

OR implements union of compatible phase masks. Intersection uses AND, subtraction uses AND with complement, and reversible changes use XOR. All may share an SDF encoding without all being the same operation. A bit shift transports an addressed pattern; a flip changes its value. The reference distinguishes both operations.

[S3, pp. 2--4; current corpus requirement that operators themselves be fields; truth-table profile introduced here]

13 Profile R-bio: the source's frontier law

RETAINED EQUATION / EXPLICIT INVARIANT AND SHORTEST-PATH RESULT

The new source calls its frontier rule “Profile R.” This revision uses the unambiguous label R_{bio} for that rule; it is separate from the earlier temporal example that also used R. [S4, p. 9]

Fix an allowed mask M and seeds $S \subseteq M$. Let $\mathcal{N}_v$ be the OR of the declared neighbor shifts. Set

$$F_0 = O_0 = S, \quad (13.1)$$

$$C_{n+1} = \mathcal{N}_v(O_n), \quad (13.2)$$

$$O_{n+1} = M \wedge \neg F_n \wedge C_{n+1}, \quad (13.3)$$

$$F_{n+1} = F_n \oplus O_{n+1} = F_n \vee O_{n+1}. \quad (13.4)$$

The last equality follows because $F_n \wedge O_{n+1} = 0$. It is not an unconditional equality between XOR and OR.

P5 / Monotonicity and first arrival

Let $\ell_S(v)$ be the shortest number of allowed graph edges from any seed. Then

$$O_n = \{v : \ell_S(v) = n\}, \quad F_n = \{v : \ell_S(v) \leq n\}. \quad (13.5)$$

At step zero, precisely the seeds have distance zero. At the next step, a newly admitted cell has a distance- n neighbor and was not reached earlier, so its distance is $n + 1$. Conversely, the penultimate vertex of any shortest path of length $n + 1$ is in O_n and causes admission. Induction proves the identities and

$$\text{supp } F_n \subseteq \text{supp } F_{n+1} \subseteq \text{supp } M. \quad (13.6)$$

For finite M , each nonempty new frontier adds a previously unvisited cell. The wave stops at its first empty frontier; with nonempty seeds, this takes at most $|\text{supp } M|$ updates including the final empty result. Empty seeds stop immediately.

What the theorem establishes

It proves non-erasure of the visited set and shortest *hop* arrival in the declared graph. It does not prove that several independently labeled waves cannot meet, that grid distance equals Euclidean distance, or that a biological composition should monotonically grow. Those are different assertions.

Readout stores frontier layers or equivalent predecessors. A shortest route is recovered by moving through adjacent cells in the immediately preceding layer. The terminal visited mask alone does not retain all predecessor choices.

14 The bug finder as a witness-producing observer

NEW CONSTRUCTION / EXAMINE THE PROPOSAL BEFORE IT IS MASKED

The source calls $F \wedge m \neq 0$ an anomaly of a supposedly fresh operator. If m is already defined as $M \wedge \neg F \wedge C$, that test is always false by construction. It cannot reveal proposals that were removed by the mask. [S4, p. 9]

The enhanced observer records the *raw proposal* P before projection:

$$P_{\text{fresh}} = P \wedge M \wedge \neg F, \quad (14.1)$$

$$P_{\text{repeat}} = P \wedge F, \quad (14.2)$$

$$P_{\text{outside}} = P \wedge \neg M. \quad (14.3)$$

Assume the current state satisfies $F \subseteq M$. Then the three masks are pairwise disjoint and

$$\boxed{P = P_{\text{fresh}} \vee P_{\text{repeat}} \vee P_{\text{outside}}.} \quad (14.4)$$

Every proposed cell is either outside M , inside F , or in $M \setminus F$. This proves the partition. The observer preserves all three classes and does not silently erase evidence of the original event.

Worked witness

On a four-cell row, let $M = 0111_2$, $F = 0001_2$ and raw proposal $P = 1011_2$. Then

$$P_{\text{fresh}} = 0010_2, \quad P_{\text{repeat}} = 0001_2, \quad P_{\text{outside}} = 1000_2. \quad (14.5)$$

The selected monotone profile returns $F' = 0011_2$. An unrestricted XOR update would give $F \oplus P = 1010_2$, erasing an old cell and setting an outside cell. Both the proposal and the derived outcomes remain available for inspection.

A repeated arrival is not always an error

A normal wave often proposes previously visited neighbors. The observer classifies the event; the profile or query states whether that class is expected, anomalous or forbidden for its task. Detection and control are separate operations.

Verified execution

Every raw proposal over every $F \subseteq M$ state on a six-cell row was checked: $6^6 = 46,656$ cases. The partition, disjointness, truth-field evaluation and update identities passed exactly.

15 Overlapping layers, ownership and contact

SEPARATE WITNESSES FOR SEPARATE MEANINGS OF OVERLAP

Layer overlap can be intentional: a chemical annotation may be associated with a chamber and a composition record at the same address. Therefore “no overlap” is not a universal criterion for a three-layer memory palace.

Let $F^{(1)}, \dots, F^{(k)}$ be labeled occupancy or ownership planes. Compute their union and their multiple-label region using

$$S_0 = 0, \quad C_0 = 0, \quad C_i = C_{i-1} \vee (S_{i-1} \wedge F^{(i)}), \quad S_i = S_{i-1} \vee F^{(i)}. \quad (15.1)$$

Induction shows that $C_k(x) = 1$ exactly when at least two labels cover x . The union alone cannot recover which labels collided; the planes or a provenance record retain that information.

Two perfectly fresh waves can meet

On the three-vertex path 0--1--2, initialize one labeled wave at 0 and another at 2. Each next frontier is $\{1\}$, and each satisfies its own freshness condition. Their intersection is nevertheless $\{1\}$.

Thus single-wave monotonicity does not prove a collision-free multi-agent or multi-material configuration. It proves the theorem in Section 13; the multiple-label observer proves a different property.

Zero confluence is a query, not a phenotype

Define a common-edge zero query by

$$Z_{\mu\chi\beta} = Z_\mu \cap Z_\chi \cap Z_{\beta,\tau}. \quad (15.2)$$

Its elements identify interfaces co-located under the same declared edge-coordinate convention. The equality does not establish hydrophobicity, adhesion, growth, equilibrium or another biological function. Those require corresponding predicates and evidence.

A record may be a valid compositional combination with $Z_{\mu\chi\beta} = \emptyset$. The worked chamber deliberately demonstrates this. Geometric co-location and catalogue membership are independent tests.

The bug-finder output is a **typed witness tuple**: repeated proposals, outside proposals, multi-label overlap, violated room predicates, undefined metric data and differing ledger assertions. None is collapsed into a single ambiguous “bug” bit.

16 CSG, offsets and smooth blending: verified meanings

SOURCE FORMULAS RETAINED WITH EXACT SET AND METRIC INTERPRETATIONS

For inside-negative scalar functions f_P, f_Q representing closed occupied sets by $\{f \leq 0\}$, the following identities are exact:

$$\{\min(f_P, f_Q) \leq 0\} = P \cup Q, \quad \{\max(f_P, f_Q) \leq 0\} = P \cap Q. \quad (16.1)$$

They are *set-membership* identities. They do not, without further conditions, guarantee exact signed-distance magnitude or identify every zero as the boundary of the final set. [S4, pp. 2, 5, 7; R1]

Two explicit counterexamples

For halfspaces $P = \{x \leq 0\}$ and $Q = \{y \leq 0\}$, take $f_P = x$ and $f_Q = y$. At $(1, 1)$, $\max(f_P, f_Q) = 1$, while the distance to the negative quadrant is $\sqrt{2}$. The combined function is not an exact SDF there.

In one dimension, $f_P(x) = x$ and $f_Q(x) = -x$ represent two closed half-lines whose union is all of $\mathbb{R}$. Yet $\min(f_P, f_Q) = -|x|$ has a zero at 0. That zero is an internal seam, not a boundary of the union. Reconstruct the final occupied geometry and its boundary when that distinction matters.

The direction of the source's padding offset

For $\delta > 0$ and an inside-negative exact SDF, $f + \delta = 0$ selects $f = -\delta$, an *inward* level. The outward level is $f - \delta = 0$. For $f(x) = |x| - 1$ and $\delta = 1/4$, the respective roots are $\pm 3/4$ and $\pm 5/4$.

A geometric separation statement needs its own condition. For example, if two closed sets have separation greater than $\delta_P + \delta_Q$, their corresponding outward metric neighborhoods cannot intersect, by the triangle inequality. Address-space isolation is still a separate property.

Smoothing changes the field

A chosen soft minimum, such as

$$\text{smin}_k(a, b) = -k^{-1} \log(e^{-ka} + e^{-kb}), \quad k > 0, \quad (16.2)$$

has value $-\log 2/k$ at $a = b = 0$. It shifts that level and is not automatically an exact SDF. The reference integer profile does not invoke this transcendental operation; an approximation would require a declared precision and reconstruction rule.

[S4, pp. 5, 7--8; counterexamples and offset direction derived here]

17 A = T, snapshots and field-time

RETAINED IDENTITY / EXECUTABLE TEMPORAL BINDINGS

The names A and T identify the same time-operator definition. In the reference package its record contains its own id, its next-state expressions and the observations attached to each step. Time is not assigned to an arbitrary material or composition channel simply because that channel occupies memory.

For the frontier profile, the temporal operation is

$$T : (M, F_n, O_n, \mathcal{L}_n) \mapsto (M, F_{n+1}, O_{n+1}, \mathcal{L}_{n+1}), \quad (17.1)$$

with the equations of R_{bio} and an explicitly recorded witness. The data and its observer description share the canonical field encoding.

Three compatible time readings

Reading	Meaning
Discrete snapshot n	The sequence position of a state transition.
Local field-time $T_{F,n}(x) = d_{F_n}(x)$	The source's SDF=T convention, under the selected normalized metric.
First-arrival $\tau(v)$	Minimum graph hops from the seed set in the fixed wave problem.

Present is $T_{F,n} = 0$ on the relevant interface. A readout called Now must identify which reading and which snapshot it selects. A zero-edge query is not silently converted into elapsed physical time. [S2, pp. 7--8; S4, p. 9]

Self-reference with an explicit evaluation boundary

A rule can quote its own canonical description, compare revisions or generate a proposed next definition. The record identity remains readable, and a temporal binding specifies when a new value becomes active. A finite description may point to itself without infinitely expanding its serialized bytes.

The reference reads the stored temporal expression graph through a common evaluator. Replacing its next-state fields with identity references pauses both state roles; this was tested without changing the evaluator. The names A and T refer to the same definition object. It does not claim that its Python bootstrap was generated from a self-hosted compiler or that all earlier corpus terms have acquired operational equations.

The stopping condition for the finite wave is its empty frontier, proved in Section 13. Entropy, biological density, a self-reference and a parity flip are not substituted for that explicit termination argument.

18 Joint effort: a provenance-preserving palace ledger

COLLABORATIVE DEFINITIONS / EVIDENCE IS RETAINED RATHER THAN OVERWRITTEN

The author's "joint effort" becomes an append-only collection of contributions. An assertion record contains

$$\ell = (\text{namespace}, \text{object}, \text{predicate}, \text{context}, \text{value}, \text{sources}). \quad (18.1)$$

A context fixes the catalogue version, geometry, units, time and relevant conditions. Assertions made under different contexts are not silently equated.

The mathematical key is the exact canonical encoding of the first four fields. Assertions with the same key and value merge their source sets. Assertions with the same key but different values retain separate variants and produce a differing-assertions witness. A SHA-256 digest in the package is a reproducibility checksum; injective room identity is proved by the indexing construction, not assumed from a hash.

P6 / Order-independent merge

The merge consists of set union on exact assertion variants and set union on their provenance. Consequently, repeating a contribution is idempotent, changing contribution order is immaterial, and grouping batches differently yields the same accumulated assertion set. No selected "winner" is needed to preserve the record.

For example, contributors A and B may record the same predicate as true, while C records it as false under the same context. Both variants remain visible, with their respective sources. The ledger does not turn disagreement into a silently overwritten Boolean truth.

Incomplete predicates

For a declared query, use {pass, fail, unknown}. Conjunction is fail if any required condition is known to fail; it is pass only when all are known to pass; otherwise it is unknown. This avoids interpreting unprovided chemical or biological data as zeros or favorable outcomes.

A verified geometric subclaim can coexist with an unknown biological effect. A room can therefore have a proven address, a tested threshold computation and an unresolved mechanism assertion without forcing those statuses into one number.

The reference ledger merge was checked for order independence and idempotence on a set containing differing assertions. The model preserves contributors' evidence; it does not manufacture experimental agreement.

[S4, author's joint-effort instruction p. 3; ledger and partial-evidence algebra introduced here]

19 Worked palace: 45 combinations, 19 predicate matches

FULLY ASSIGNED SYNTHETIC CATALOGUE / EXACT BIT-PARALLEL SELECTION

The example deliberately uses synthetic labels and dimensions. It tests the representation and rules rather than asserting biological efficacy.

All chamber and annotation masks are on an 8×8 grid. Ranges below include both endpoints. The microstructure role is cavity.

Catalogue	Geometry / counts	Declared parameter
μ_0	$x = 1 \dots 6, y = 1 \dots 6$	Count limit 4
μ_1	$x = 2 \dots 5, y = 1 \dots 6$	Count limit 2
μ_2	$x = 1 \dots 6, y = 2 \dots 5$	Count limit 3
χ_0	$x = 2 \dots 3, y = 2 \dots 3$	Count minimum 0
χ_1	$x = 1 \dots 4, y = 1 \dots 4$	Count minimum 1
χ_2	$x = 4 \dots 6, y = 4 \dots 6$	Count minimum 2
$\beta_b, b = 0 \dots 4$	Numerators ($b, 4 - b$), denominator 4	Anonymous components

A room passes three *declared bookkeeping predicates*: the annotation lies inside its chamber; its first component count does not exceed that chamber's assigned limit; and that count reaches the annotation's assigned minimum. These numerical limits are selected example data, not fitted biological laws.

The candidate count is $3 \cdot 3 \cdot 5 = 45$. Each predicate yields a 45-bit field. ALL_THREE intersects those fields. Passing-room counts by catalogue pair are:

	χ_0	χ_1	χ_2	Total
μ_0	5	4	3	12
μ_1	3	0	0	3
μ_2	4	0	0	4
Total	12	4	3	19

The resulting packed room-selection integer is 16106386399. The code checks every selection bit against its scalar room record and retains all failure witnesses. A passing-room bit is not a judgment that a real chemical/bacterial combination is viable.

20 One room: zero-confluence, composition and readout

CALCULATED EXAMPLE / ALL QUANTITIES CAN BE REPRODUCED

Select (μ_0, χ_0, β_2) . Its index is

$$q = ((0 \cdot 3) + 0) \cdot 5 + 2 = 2. \quad (20.1)$$

The composition record is $(2, 2)/4 = (1/2, 1/2)$. The annotation is a 2×2 square inside a 6×6 chamber, so all three declared predicates pass.

For illustration, define the composition's occupied support as the chamber/annotation intersection. In this room it equals the annotation. Hence $Z_\chi = Z_\beta$, but the chamber's boundary lies farther out:

$$Z_\mu \cap Z_\chi \cap Z_\beta = \emptyset. \quad (20.2)$$

This gives a fully specified valid catalogue record without a triple-zero confluence. The source's proposed return vector $[0, 0, 0]$ is a possible query result, not a necessary condition for every mechanism record. [S4, p. 8]

Route through the chamber

Use $M = F_{\mu_0}$ and seed $(1, 1)$. The target is $(6, 6)$. The layer sizes are

$$1, 2, 3, 4, 5, 6, 5, 4, 3, 2, 1.$$

Their sum is 36, the chamber cell count. The last nonempty frontier is layer 10; update 11 returns the empty frontier. The recovered route is

$$\begin{aligned} (1, 1) &\rightarrow (2, 1) \rightarrow (3, 1) \rightarrow (4, 1) \rightarrow (5, 1) \rightarrow (6, 1) \\ &\rightarrow (6, 2) \rightarrow (6, 3) \rightarrow (6, 4) \rightarrow (6, 5) \rightarrow (6, 6). \end{aligned}$$

It uses 10 graph edges. The Manhattan lower bound is also $|6 - 1| + |6 - 1| = 10$, establishing optimality directly as well as through the frontier theorem.

What is returned

The readout includes the room tuple and version, typed layer records, component numerators/denominator, predicate results and witnesses, zero-confluence edge count, route and frontier statistics. It also retains the exact rule and time definitions used.

ELI5

A room is a labeled combination, not a claim that all its contents fuse into one organism. You can ask whether its labels match your rules, whether its regions touch, and how to move through its geometry. Each question returns its own answer and evidence.

21 Executable reference and reproducibility

DELIVERED PACKAGE / INTEGER AND RATIONAL ARITHMETIC

The companion ZIP contains the formalization, Python reference, examples, source ledger and complete verification record. It uses Python 3.10 or later and the standard library only. The computational masks, shifts, counts, addresses and thresholds are integer-based; normalized checks use exact rational numbers.

Run from the extracted package directory:

```
python src/palace.py --output examples/reproduced_demo.json
python tests/verify.py --json verification/reproduced_tests.json
```

The first command builds the 45-room example and reports its witnesses. The second executes the finite verification suites. Tests use explicit assertions and should be run without Python's optimization switch that removes assertions.

Core update, with the observation kept separate

```
raw = grid.neighbours(operator)
fresh = truth_field(rules['FRESH'], raw, allowed, field, full)
repeated = truth_field(rules['AND'], raw, field, 0, full)
outside = truth_field(rules['AND_NOT'], raw, allowed, 0, full)
next_field = truth_field(rules['OR'], field, fresh, 0, full)
return raw, fresh, repeated, outside, next_field, fresh
```

The `proposal` argument can also supply a raw mask directly, allowing the observer to inspect deliberate test events. Witnesses are returned as data. There is no implicit suppression of a source assertion or hidden rule that changes its provenance.

Scope of the delivered execution

The reference tests a finite two-dimensional grid realization, exact descriptor encodings and combinatorial records. The formal schema permits three-dimensional carriers and richer component catalogues when they are assigned. No GPU timing, growth rate, biological effect or material-performance result is attached to the synthetic examples.

The previous project's GPU/texture direction remains compatible with these bitplanes, but this revision does not claim a new CUDA build or cache benchmark. The delivered artifact is the enhanced formal specification plus a runnable verification reference for its new content.

22 Verification ledger

EXECUTED SUITES / COUNTS KEPT SEPARATE FROM THEOREM SCOPE

The reference implementation was executed for this revision and returned PASS. The following are explicit test categories; they are not added together as though they were independent empirical studies.

Verification category	Count
Exhaustive 3×3 allowed-mask/seed configurations	19,683
Decoded target results within those configurations	177,147
Exact graph-distance fields	4,608
Vertex distance/complement checks within those fields	53,760
Raw-proposal partition configurations	46,656
Truth-table row checks	2,048
Data-defined time evaluations, including paused rules	2,592
Canonical definition-SDF samples	43,008
Mixed-radix room-index round trips	3,375
Octree/Morton coordinate round trips	37,449
Quantized count-threshold cases	36,864
Exact three-component rational compositions	164
Three-layer pack/unpack round trips	500
Multiple-label overlap patterns	4,096
Conservative integer-transfer cases	2,000
Collaborative ledger checks	12
Partial-evidence algebra cases	27

Independent comparisons

Packed wave arrivals are compared with coordinate-based queue traversal. Distance values are compared with direct minimum distances to all candidate midpoint boundaries. The room selector is compared bit by bit with individual room predicates. Ledger variants are compared across contribution orders. All arithmetic is exact in the tested domain.

Proof versus test

The finite tests exercise implementations. The proofs establish general statements under their explicit hypotheses: canonical SDF exactness, graph-distance reconstruction, conservation, address injectivity, wave first-arrival correctness, proposal partition and order-independent merge.

Counterexamples in Sections 6, 7, 15 and 16 show why several broad claims in the exported explanation need narrower mathematical meanings. They are part of verification rather than assertions that the requested framework cannot be developed.

23 Sources, retained parameters and consolidated definition

TRACEABILITY / WHAT THE ENHANCED CORPUS NOW DEFINES

S1. Original six-page source beginning *men T room a circle a sphere* Vocabulary on pages 1--3; author's ALL-operators Axis 0 statement on page 5.

S2. Eight-page hyphenated-filename continuation. Operator/field closure on pages 4--5; seed/readout and SDF=T, Present and Now on pages 6--8.

S3. Six-page ladybug/Zaandam source. Author's Boolean double UU instruction on page 2; two-layer interpretation and shift/flip discussion on pages 3--4.

S4. Current PDF beginning *bromofly and derivatives in nature insects bug finder 0 day signed distance field notations*. Nine content pages plus a blank final page. Author correction: page 3. Rooms/corridors table: pages 4--5. Three-layer pipeline: page 6. Coupling proposals: pages 7--8. Corpus reinterpretation: page 9.

R1. John C. Hart, "Sphere tracing: a geometric method for the antialiased ray tracing of implicit surfaces," *The Visual Computer* 12 (1996), 527--545. DOI: 10.1007/s003710050084. Section 2.1 distinguishes distance functions and bounds; Section 2.4 discusses CSG. [Paper consulted](#). Used as mathematical background, not as an instruction to build a renderer.

R2. IUPAC Gold Book, [molecular graph](#), DOI: 10.1351/goldbook.MT07069. Definition used for labeled atom/bond graph representation. External sources consulted 18 September 2026.

Parameters that a new room must actually provide

A physical metric or reconstruction, layer role and units, chemical graph/geometry data, composition identities/counts, interaction predicates, temporal law and evidence sources. Source names not assigned an evaluation law remain named parameters; their meanings are not guessed.

Consolidated v6 definition

Axis 0 / the zero-boundary combinatorial memory palace is a common operator-field representation with a typed, versioned product of microstructure, chemical-geometry and composition records. Its room identities are injective, its zero queries have declared metrics, its temporal profiles have explicit rules, and its observer returns inspectable witnesses. Source assertions, constructed models, mathematical proofs, executed results and empirical observations remain linked but distinguishable.